

OLIST CUSTOMER SATISFACTION ANALYTICS

An End-to-End E-commerce Analytics Case Study: Investigating the operational drivers of customer satisfaction and translating evidence into business action

Tools: SQL Server | Tableau

Analysis: Data Validation | Exploratory Analysis | Business Intelligence | Data Visualisation

Dataset: Olist Brazilian E-commerce Public Dataset

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Case Study Overview

This case study investigates the operational and commercial factors associated with customer satisfaction within the Olist e-commerce marketplace.

The project follows the complete analytical journey from defining the business problem and understanding the underlying data, through data-quality validation and SQL analysis, to the development of an executive Tableau dashboard and evidence-based business recommendations.

Rather than beginning with a dashboard and searching for a story afterwards, the analysis follows a business-first approach:

Business Problem → Data Understanding → Data Validation → Evidence → Insight → Decision Support

Case Study Structure: Table

Chapter	Focus
Chapter 1	Defining the Business Problem
Chapter 2	Understanding the Data Landscape
Chapter 3	Exploring and Validating the Evidence
Chapter 4	Developing the Insights
Chapter 5	From Insight to Business Action

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Chapter 1: Defining the Business Problem

1.1 Business Context

Olist operates a Brazilian e-commerce marketplace connecting customers with independent sellers. In this environment, customer satisfaction is influenced by more than the product purchased. From the customer's perspective, the seller, delivery experience, product and marketplace form one connected purchasing journey.

A review score therefore tells the business how a customer felt about an experience, but not necessarily what created that experience. A low rating could be associated with late delivery, seller performance, particular areas of the product catalogue or a combination of factors.

This case study investigates those relationships using the Olist e-commerce dataset. The goal is to move beyond reporting an overall customer rating and identify operational and commercial patterns that could help Olist decide where customer-experience problems are concentrated and where investigation should begin.

Business Problem

Olist needs to understand which operational and commercial factors are associated with customer satisfaction, where dissatisfaction is concentrated, and which areas should receive the greatest attention.

The project was therefore designed around a simple shift in thinking:

Instead of asking...	The analysis asks...
What is the average review score?	What sits behind that score?
How many customers are dissatisfied?	Where is dissatisfaction concentrated?
Which metric looks worst?	Which problems have meaningful customer impact?
What happened?	Where should the business investigate first?

1.2 Why This Matters

Customer satisfaction is not only a reporting metric. In a marketplace, it can also act as a signal that something within the customer journey is not performing as expected.

Looking only at a marketplace-wide average creates a particular risk: important problems can disappear inside an otherwise healthy number. A product category, seller or delivery segment may perform considerably worse than the marketplace overall without being obvious at executive level.

There is also a prioritisation problem. Olist cannot investigate every seller, category and order simultaneously. The business needs evidence showing which problems are both significant and large enough to justify attention.

If the Business Does Not Improve

Without a structured way to identify and prioritise dissatisfaction, Olist could:

- overlook recurring delivery problems because they appear as individual incidents;
- miss underperforming sellers hidden within marketplace averages;
- focus on categories with dramatic percentages but limited customer impact;
- overlook larger categories generating substantially more negative experiences;
- spend operational resources on the most visible problems rather than the most meaningful ones; and
- struggle to measure whether future interventions actually improve customer experience.

The dataset does **not** allow this case study to calculate the financial cost of dissatisfaction or prove its effect on customer retention. I therefore do not assign an invented monetary value to these risks. Instead, the analysis focuses on what the available evidence can support: identifying operational patterns associated with poorer customer experiences and establishing where the business should investigate further.

1.3 Analytical Objective

The objective of the project is Identify operational and commercial patterns associated with customer satisfaction and translate those findings into a practical framework for prioritising customer-experience improvement.

To achieve this, I divided the investigation into three analytical areas:

Area	Business Question	Why It Matters
Delivery Performance	How is delivery performance associated with customer satisfaction?	Determines whether delivery reliability should become a customer-experience priority.
Product Categories	Where is dissatisfaction concentrated across product categories?	Helps distinguish high percentages from problems affecting meaningful numbers of customers.
Seller Performance	Which sellers show signs of customer or operational underperformance?	Help identify marketplace partners requiring monitoring or deeper investigation.

These questions created the analytical direction for the rest of the project.

1.4 Stakeholders and Decisions

The analysis was designed to support different levels of decision-making rather than produce one generic report.

Stakeholder	What They Need From the Analysis
Operations	Identify delivery-performance problems and areas requiring operational investigation.
Seller / Marketplace Management	Identify sellers showing persistent customer or fulfilment risk.
Commercial / Category Teams	Understand where dissatisfaction affects meaningful transaction volumes.
Leadership	Understand the strongest customer-experience risks and where intervention should be prioritised.

1.5 Proposed Analytical Solution

The solution was designed to connect customer reviews with the operational context surrounding each order.

The investigation would examine three relationships:

Customer Reviews + Delivery Performance

To determine whether satisfaction changes as delivery performance worsens.

Customer Reviews + Product Categories

To identify categories where dissatisfaction is elevated and determine whether that risk occurs at meaningful scale.

Customer Reviews + Seller Performance

To identify sellers where customer feedback and operational performance suggest the need for closer investigation.

Importantly, the project would not move directly from these relationships to recommendations. The data first needed to be understood and validated.

The full analytical process therefore became:

Business Problem → Data Understanding → Data Validation → Evidence → Insight → Decision Support

1.6 What Success Looks Like

The success of this project is not measured by the number of SQL queries written or charts created. A useful analysis should leave a stakeholder able to answer four questions:

1. **What is happening with customer satisfaction?**
2. **Where is dissatisfaction concentrated?**
3. **Which operational patterns are associated with poorer experiences?**
4. **Where should the business investigate or intervene first?**

If the final analysis cannot support those decisions, then it has produced information rather than useful decision support.

1.7 Analytical Principle

One rule guided the investigation from the beginning:

No conclusion would be made until it was supported by evidence from the data.

This meant avoiding convenient assumptions. The relationship between late delivery and low review scores would not automatically be described as causal. A category with the highest negative-review percentage would not automatically become the largest business priority. A seller would not be classified as underperforming based on one metric or a handful of orders.

The objective was to allow the evidence to determine the findings rather than search for evidence that supported an expected conclusion.

Chapter 1 Outcome

Chapter 1 established the business case for the analysis and narrowed the investigation to three areas: **delivery performance, product-category risk and seller performance.**

With the business questions defined, the next step was to determine whether the available Olist data could reliably support the investigation. This required understanding the datasets, their relationships and the measures needed for analysis.

Chapter 2: Understanding the Data Landscape

2.1 Why This Stage Matters

Before investigating what influences customer satisfaction, I needed to understand how the Olist data represented the customer journey.

The dataset is relational rather than a single analysis-ready table. Information about an order is distributed across separate tables for customers, products, sellers, payments, reviews, order items and delivery activity. This means answering even a relatively simple business question may require information from several tables.

That creates analytical risk. If tables are joined at the wrong level, one order can appear multiple times and distort metrics such as average review score, order count or revenue. Similarly, calculating a seller metric at product level or an order metric at item level could produce results that look reasonable while being incorrect.

For that reason, Chapter 2 focused on three things before any conclusions were drawn:

- understanding what each dataset represents;
- identifying how the datasets connect; and
- defining the measures required to answer the business questions established in Chapter 1.

The purpose was to create a reliable analytical foundation before moving into data validation and SQL exploration.

2.2 The Olist Data Environment

The project uses eight core datasets, each representing a different part of the marketplace.

Dataset	Records	Role in the Analysis
Orders	99,441	Order lifecycle, purchase dates, order status and delivery performance
Order Items	112,650	Connects orders to products and sellers and contains item-level price and freight information
Customers	99,441	Connects each order to customer information
Products	32,951	Product attributes and product-category information
Sellers	3,095	Seller identity and location

Order Reviews	99,224	Customer review scores used as the primary satisfaction measure
Order Payments	103,886	Payment values and payment information
Geolocation	1,000,163	Geographic reference information

These row counts immediately revealed something important about the structure of the data: **the tables do not all operate at the same level of detail.**

For example, there are more order-item records than orders because a single order can contain multiple items. There are also more payment records than orders because an order may involve more than one payment record.

Recognising these differences became important when deciding how the tables should be joined later.

2.3 Understanding the Grain of the Data

One of the most important decisions in this project was identifying the **grain**, or what one row represents, in each table.

Table	Approximate Grain
Orders	One row per order
Customers	Customer record associated with an order
Order Items	One row per item within an order
Products	One row per product
Sellers	One row per seller
Reviews	Review record associated with an order
Payments	One row per payment transaction or payment sequence
Geolocation	Geographic reference record

This distinction matters because joining a one-row-per-order table to a one-row-per-item table changes the number of rows in the result.

For example, if an order contains three products, joining `orders` to `order_items` can produce three rows for that order. A simple `COUNT(order_id)` after that join could therefore count the same order multiple times.

For order-level metrics, I needed to use the appropriate level of aggregation, including techniques such as `COUNT(DISTINCT order_id)` where required.

This was not just a SQL consideration. It affected whether the business metrics produced later could be trusted.

2.4 Mapping the Customer Journey Through the Database

Once the role and grain of each dataset were understood, I mapped how they could be used to reconstruct the customer experience.

The **orders table** became the central analytical table because it contains the order lifecycle and connects the transaction to the customer.

From there, the analysis could follow several paths.

Customer satisfaction path

Order → Review

This connects the transaction to the review score eventually left by the customer.

Delivery path

Order → Purchase Date → Estimated Delivery Date → Actual Delivery Date → Review

This provides the information needed to compare promised and actual delivery performance with customer satisfaction.

Product category path

Order → Order Item → Product → Product Category → Review

This allows customer feedback to be examined across different areas of the product catalogue.

Seller path

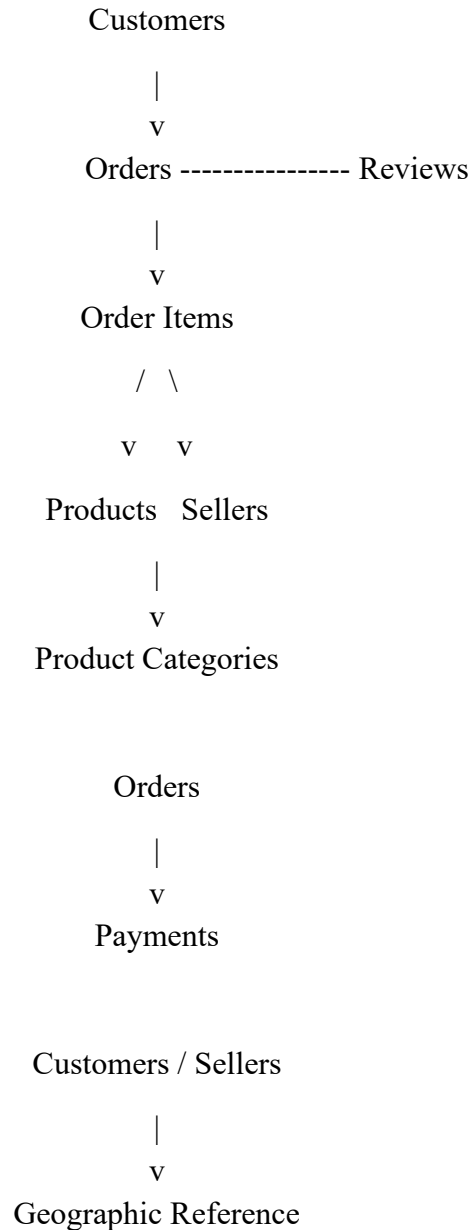
Order → Order Item → Seller → Review

This allows seller-level operational and customer-satisfaction performance to be investigated.

Together, these relationships created the analytical structure needed to answer the three business questions defined in Chapter 1.

2.5 Relationship Map

The working relationship structure can be summarised as:



The important part of this model was not simply identifying foreign keys. It was understanding **which relationships were relevant to each business question**.

For example, geolocation data was available, but the strongest customer-satisfaction questions did not require geographic analysis during the main investigation. I therefore avoided adding geographic complexity simply because the data existed.

That decision kept the project focused on the business problem rather than trying to use every available field.

2.6 Connecting the Data to the Business Questions

With the relationships mapped, I could determine exactly which datasets were required for each part of the investigation.

Business Question	Main Data Required	Key Measures
How is delivery performance associated with customer satisfaction?	Orders + Reviews	Delivery timing, review score, order count
Which product categories show greater customer dissatisfaction?	Orders + Order Items + Products + Reviews	Category, review score, negative reviews, order volume
Which sellers show signs of underperformance?	Orders + Order Items + Sellers + Reviews	Seller, order volume, review score, negative reviews, late-delivery rate
What is the overall marketplace context?	Orders + Reviews + Order Items / Payments where appropriate	Delivered orders, review score, GMV, late-delivery rate, negative-review rate, freight cost

This step was useful because it prevented unnecessary joins.

Rather than building one enormous table containing every available dataset, each analytical question could use only the tables required to answer it.

That reduces complexity and makes the SQL easier to validate.

2.7 Defining Customer Satisfaction

The review_score field became the primary measure of customer satisfaction.

Olist reviews use a five-point scale, allowing the analysis to examine both average satisfaction and the distribution of poor customer experiences.

Two complementary measures were therefore planned:

Average Review Score

This provides an overall indication of satisfaction within a group of orders.

Negative Reviews

Reviews with a score of **1 or 2** were treated as negative experiences for the purposes of this case study.

Using both measures was important.

The average can summarise overall performance, but it can hide the number of customers having particularly poor experiences. Negative-review rate provides another way of identifying where dissatisfaction is concentrated.

Neither measure was intended to explain *why* a customer was dissatisfied on its own. They provided the outcome measures that could later be compared with operational characteristics.

2.8 KPI Framework

Before beginning the main analysis, I established a KPI framework so that measures would be defined consistently throughout the project.

KPI Area	Measures	Purpose
Customer Experience	Average review score, negative-review count, negative-review rate	Measure overall satisfaction and dissatisfaction
Delivery Performance	Delivery timing, late-delivery rate	Evaluate reliability against estimated delivery dates
Marketplace Scale	Delivered orders, order volume	Provide context for the size of findings
Commercial Performance	GMV, freight cost	Add commercial context to marketplace activity
Category Performance	Order volume, average review score, negative reviews	Compare dissatisfaction across product categories
Seller Performance	Order volume, review score, negative reviews, late-delivery rate, revenue	Evaluate seller performance using multiple indicators

This framework later became especially useful when comparing product categories and sellers.

A poor percentage alone does not necessarily represent the greatest business risk. Volume provides context about how many transactions are affected.

That idea would become one of the most important analytical principles in Chapter 4:

Performance should be evaluated using both severity and scale.

2.9 Analytical Scope

Understanding the data also helped define what the project would **not** attempt to answer.

The dataset contains many possible analytical directions, including geography, payment behaviour and detailed product characteristics. Those areas may be useful for separate investigations, but including all of them would have weakened the focus of this case study.

The scope remained centered on the three areas established in Chapter 1:

- delivery performance;
- product-category risk; and
- seller performance.

Payment, freight and commercial measures were retained where they provided useful marketplace context, while other available fields were not forced into the analysis simply because they existed.

This is an important part of analytical decision-making. A strong analysis is not necessarily the one that uses the most data. It is the one that uses the **right data for the decision being investigated**.

2.10 Why the Data Model Matters to the Final Solution

The solution proposed in Chapter 1 depended on being able to move from an overall customer review to the operational context behind it.

Chapter 2 established that the Olist relational structure makes this possible.

Reviews can be connected to individual orders. Orders contain the timestamps needed to evaluate delivery reliability. Order items connect those experiences to products and sellers. Products provide category information, while seller records allow marketplace-partner performance to be investigated separately.

This means the business problem can be examined at several useful levels:

Level	Question Supported
Marketplace	What does overall customer experience look like?
Delivery	Does satisfaction differ with delivery performance?
Category	Where is dissatisfaction concentrated across the catalogue?
Seller	Which marketplace partners require closer investigation?

Chapter 2 Outcome

Chapter 2 established the analytical foundation for the project. The eight core datasets were understood in terms of their business role, level of detail and relationship to the customer journey. The analysis also established the KPI framework and identified the specific tables required for each business question.

Most importantly, this stage highlighted a key analytical risk: **having the necessary data does not automatically mean the data is ready to be trusted**.

Before comparing delivery performance, categories or sellers, I still needed to confirm that the database was complete enough, that important fields behaved as expected and that missing values or import issues would not undermine the analysis.

Chapter 3: Exploring and Validating the Evidence

3.1 From Understanding the Data to Trusting It

Chapter 2 established how the Olist datasets connect, and which tables were required to investigate customer satisfaction. Before using those relationships to answer the business questions, I needed to determine whether the underlying data was reliable enough to support the analysis.

This stage was deliberately completed before developing business insights. A technically correct query can still produce a misleading conclusion if the source data is incomplete, incorrectly imported or misunderstood.

The validation process focused on five questions:

Validation Question	Why It Matters
Were all core datasets imported successfully?	Missing or truncated records could distort later analysis.
Are order statuses and dates internally consistent?	Delivery analysis depends on correctly understanding the order lifecycle.
Can missing values be explained?	Not every NULL represents a data-quality failure.
Is review coverage sufficient?	Customer satisfaction analysis depends on the availability of review data.
Are product categories sufficiently complete?	Missing categories could affect category-level comparisons.

This chapter therefore represents the point where the project moved from **understanding the database structure to testing whether the evidence could be trusted**.

3.2 Building the SQL Database

The source data was provided across separate Olist datasets and imported into SQL Server to create the working analytical database.

This was not completely frictionless.

During the import process, I encountered datatype errors because some of the initially assigned SQL datatypes were too restrictive for the values contained in the source files. For example, fields initially assigned smaller integer types required conversion to larger integer types such as INT so the complete range of values could be loaded successfully.

Rather than treating the import as a mechanical step, I reviewed the affected fields, adjusted the datatype definitions and reran the imports.

Why This Matters

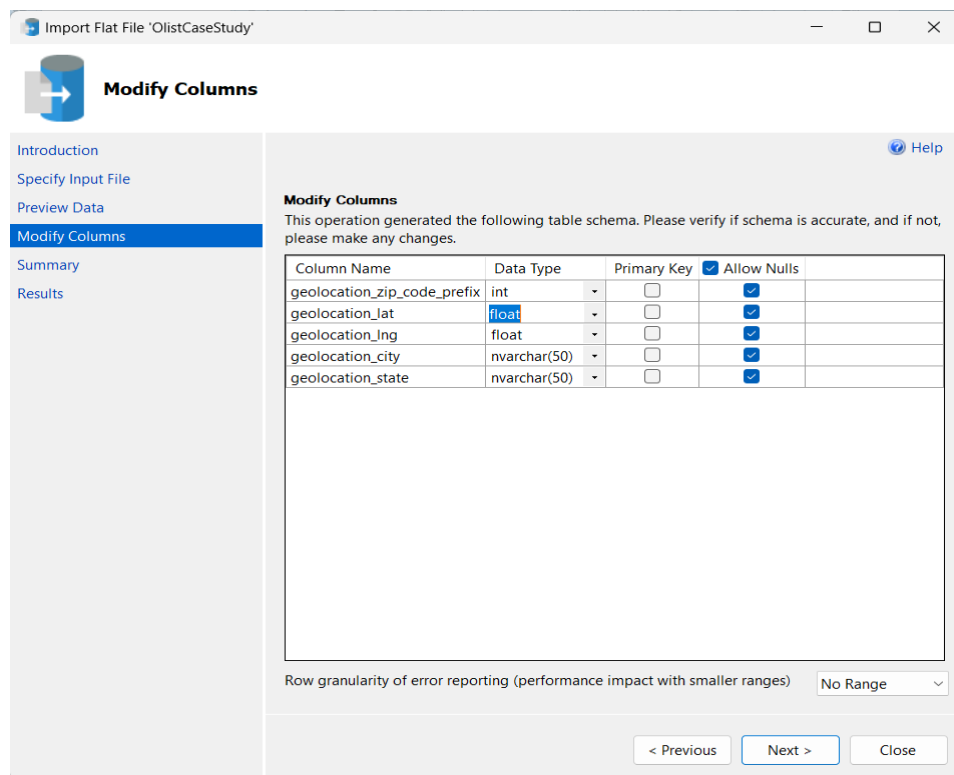
An import that completes incorrectly can create problems that are difficult to notice later. Values may fail to load, records may be rejected or a field may no longer represent the source data accurately.

For this project, resolving datatype issues before analysis meant that later SQL results were being calculated from the intended source records rather than from a partially imported database.

It also reinforced an important part of practical analytics work:

Data preparation is not separate from analytical quality. Decisions made while loading and structuring data can directly affect the reliability of every result that follows.

Evidence to Include



Import Flat File "OlistCaseStudy"

Modify Columns

Introduction
Specify Input File
Preview Data
Modify Columns
Summary
Results

Modify Columns
This operation generated the following table schema. Please verify if schema is accurate, and if not, please make any changes.

Column Name	Data Type	Primary Key	Allow Nulls
geolocation_zip_code_prefix	int	☐	☒
geolocation_lat	float	☐	☒
geolocation_lng	float	☐	☒
geolocation_city	nvarchar(50)	☐	☒
geolocation_state	nvarchar(50)	☐	☒

Row granularity of error reporting (performance impact with smaller ranges) No Range

< Previous Next > Close

Figure 1. SQL Server datatype issue encountered during the Olist database import and corrected before analysis.

3.3 Validating the Core Tables

After completing the imports, I checked the number of records in each core table.

Table	Row Count
olist_orders	99,441
olist_order_items	112,650
olist_customers	99,441
olist_products	32,951
olist_sellers	3,095
olist_order_reviews	99,224
olist_order_payments	103,886
olist_geolocation	1,000,163

These counts also supported what had been identified about table grain in Chapter 2. For example, order items exceed the number of orders because an order can contain multiple items, while payment records exceed orders because transactions can contain multiple payment records.

The validation was therefore not based on expecting every table to contain the same number of rows. The purpose was to confirm that the differences made sense given the business role of each table.

Why This Matters

Later calculations combine information across several of these datasets. Establishing the basic size and structure of each table provided an initial check that the database had been loaded successfully and helped identify where one-to-many relationships needed to be handled carefully.

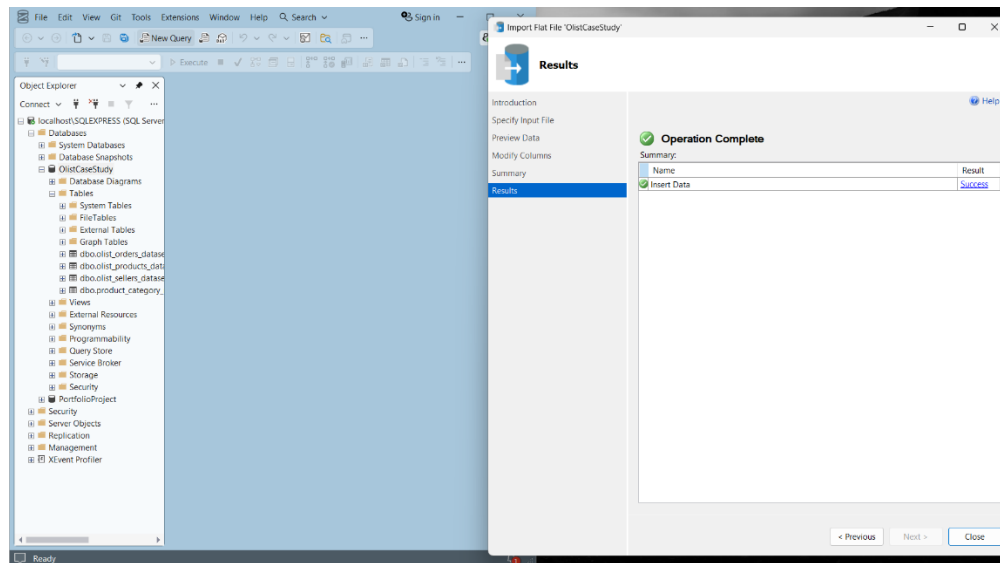


Figure 2. Row-count validation across the eight core Olist tables after database import.

3.4 Understanding the Order Population

The next step was to understand what had actually happened to the 99,441 orders in the database.

Order Status	Orders	% of Total
Delivered	96,478	97.02%
Shipped	1,107	1.11%
Canceled	625	0.63%
Unavailable	609	0.61%
Invoiced	314	0.32%
Processing	301	0.30%
Created	5	0.01%
Approved	2	<0.01%

The most important finding at this stage was that **96,478 orders, approximately 97% of the dataset, were recorded as delivered.**

That gave the project a substantial population of completed transactions from which customer experience could later be investigated.

It also showed why order status needed to remain part of the analytical logic. Orders that were cancelled, unavailable, still processing or still in transit should not automatically be treated as completed customer journeys.

Connection to the Final Solution

The final analysis focuses heavily on delivery performance and customer reviews. Establishing the delivered-order population here created the foundation for those later comparisons and prevented unfinished orders from being interpreted as completed delivery experiences.

3.5 Confirming the Historical Coverage

I also checked the earliest and latest purchase timestamps in the orders table.

Earliest Order	Latest Order	Historical Span
4 September 2016	17 October 2018	773 days

The database therefore represents a little over two years of historical marketplace activity.

Why This Matters

This establishes the time boundary of the case study.

The findings describe patterns present in the historical Olist dataset. They should **not** be presented as evidence of Olist's current operational performance.

This distinction becomes especially important when presenting the project publicly. The purpose of the case study is to demonstrate the analytical process and what can be learned from the dataset, not to make claims about how the company performs today.

3.6 Investigating Missing Delivery Dates

One of the more useful validation exercises involved missing delivery dates.

Across all **99,441 orders**:

Validation Measure	Result
Missing actual delivery date	2,965
Missing estimated delivery date	0
Missing purchase date	0
% missing actual delivery date	2.98%

At first glance, 2,965 missing actual-delivery timestamps could appear to be a data-quality problem.

Instead of immediately removing those records or filling the missing values, I investigated the order statuses associated with them.

The missing delivery dates were distributed as follows:

Order Status	Missing Actual Delivery Dates
Shipped	1,107
Canceled	619
Unavailable	609
Invoiced	314
Processing	301
Delivered	8
Created	5
Approved	2

This changed the interpretation considerably.

Most of the missing values were attached to orders that had **not reached completed delivery status**. A shipped order, for example, would not necessarily be expected to contain an actual customer delivery timestamp yet. Similarly, cancelled and unavailable orders may never have reached delivery.

Only **8 records classified as delivered** were missing the actual delivery timestamp.

Why This Matters

This was an important analytical lesson in the project:

A missing value is not automatically an error. Its meaning depends on the business process that produced it.

Simply reporting "2.98% of delivery dates are missing" would have overstated the quality problem.

By investigating order status, the missingness could largely be explained by the order lifecycle itself.

For the later delivery analysis, delivered orders without the required delivery timestamp could be excluded where that timestamp was necessary for calculating delivery performance.

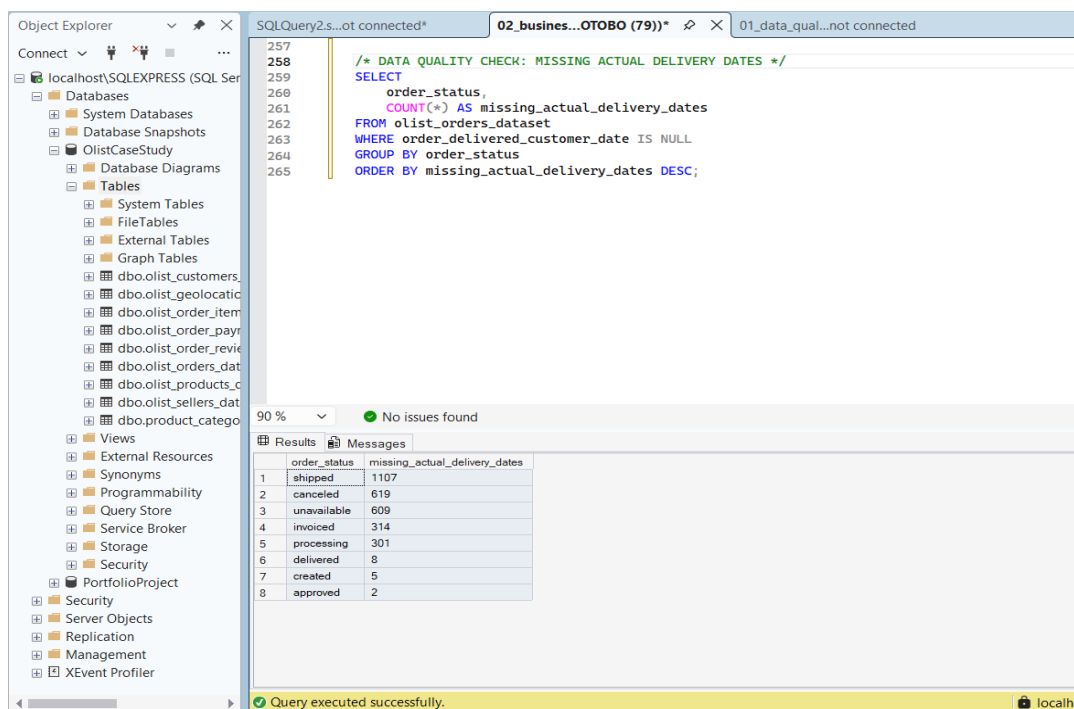


Figure 3. Investigation of missing actual-delivery dates by order status.

3.7 Validating Customer Review Data

Because customer satisfaction is the central outcome of the case study, I also examined the distribution of review scores.

Review Score	Reviews	% of Reviews
1	11,424	11.51%
2	3,151	3.18%

Overall sentiment in the dataset is positive, with more than half of recorded reviews receiving the maximum score of five.

However, the distribution also contains a meaningful population of poor experiences. Scores of 1 and 2 were therefore later used to define **negative reviews** for the purposes of the analysis.

This gave the project two complementary ways of measuring satisfaction:

- **average review score**, to understand overall performance; and
- **negative-review rate**, to understand the concentration of particularly poor experiences.

Using both became important later because averages alone can hide the number of customers experiencing serious dissatisfaction.

3.8 Checking Review Coverage

The presence of a reviews table does not mean every completed order has a review, so I checked review coverage specifically among delivered orders.

Measure	Orders
Delivered orders checked	96,478
Delivered orders with reviews	95,832
Delivered orders without reviews	646

The large majority of delivered orders therefore had associated review information, giving the analysis strong coverage for investigating customer satisfaction.

The 646 delivered orders without reviews were not treated as negative or positive experiences. They simply did not contain the outcome measure required for review-based analysis.

Why This Matters

This prevents a subtle analytical mistake.

An order without a review is **not equivalent to a satisfied customer**, and it is also not equivalent to a dissatisfied customer. Treating missing reviews as either would introduce an assumption unsupported by the data.

Later satisfaction analyses therefore use orders where the relevant review information is available.

3.9 Assessing Product Category Completeness

Product-category information was particularly important because one of the business questions concerned differences in satisfaction across categories.

The product table contained:

Measure	Result
Total products	32,951
Products missing category	610
% missing category	1.85%

Category information was therefore available for the substantial majority of products.

The missing 1.85% was documented rather than silently ignored. Where category information was unavailable during later analysis, those observations could be retained as an **Unknown** category where appropriate rather than assigning them to a category without evidence.

Why This Matters

The missing-category rate was small enough for category analysis to remain useful, but documenting it prevents the final results from implying perfect product classification.

This limitation also carries forward into the final recommendations.

3.10 What I Changed and What I Did Not Change

An important part of validation was deciding when intervention was actually necessary.

Changes made

- SQL datatypes were adjusted where the initial datatype was too restrictive for the source values.
- Analytical queries were designed to respect the grain of the underlying tables.
- Delivered orders without required delivery timestamps were excluded from calculations that depended on those timestamps.
- Missing product categories were recognised explicitly rather than assigned invented values.

Changes deliberately avoided

- Missing delivery dates were not automatically filled.
- Missing reviews were not converted into assumed review scores.
- Unknown product categories were not guessed.
- Non-delivered orders were not treated as completed deliveries.
- Valid extreme values were not removed simply because they appeared unusual.

This distinction matters because data cleaning should not mean changing every inconvenient value. The goal is to correct genuine data problems while preserving valid information produced by the underlying business process.

3.11 Validation Decision

After completing these checks, I considered whether the identified issues were severe enough to prevent the customer-satisfaction investigation from continuing.

They were not.

The dataset contained a large delivered-order population, high review coverage, largely complete product-category information and sufficient delivery information for the main analytical questions.

The limitations that remained were identifiable and could be handled explicitly in later queries.

Validation Summary

Area	Finding	Decision
Database import	Datatype issues encountered and corrected	Proceed
Core tables	Expected relational differences confirmed	Proceed
Order population	97.02% recorded as delivered	Proceed
Date coverage	773-day historical period established	Proceed with historical-context limitation
Missing delivery dates	Mostly explained by non-delivered statuses	Restrict delivery analysis appropriately
Review coverage	95,832 of 96,478 delivered orders had reviews	Proceed
Product categories	1.85% missing category information	Proceed and document limitation

The decision was therefore: The Olist dataset is sufficiently complete and internally consistent for the planned customer-satisfaction analysis, provided the documented limitations and filtering rules are respected.

3.12 How Chapter 3 Supports the Business Solution

Chapter 1 established that Olist needed a way to identify and prioritise customer-experience risk.

Chapter 2 showed how the relational data could connect reviews with delivery, categories and sellers.

Chapter 3 established that those relationships were reliable enough to investigate.

This validation matters to the final solution because the eventual dashboard and recommendations depend on comparisons between different groups of orders. If the underlying order population, reviews or delivery dates had not been checked first, a visually convincing dashboard could still have been built on unreliable assumptions.

The project was now ready to move beyond:

"Can this evidence be trusted?"

and begin answering:

"What does this evidence actually tell us about customer satisfaction?"

That became the purpose of **Chapter 4: Developing the Insights**.

Chapter 4: Developing the Insights

4.1 Moving From Validation to Investigation

By the end of Chapter 3, the dataset had been profiled, validated and considered suitable for analysis. The next stage was to return to the original business problem: **what operational and commercial patterns are associated with customer satisfaction, and where should Olist focus its attention?**

The analysis followed the three areas established in Chapter 1:

Analytical Area	Question
Delivery Performance	Does customer satisfaction change as delivery performance worsens?
Product Category Risk	Where is dissatisfaction concentrated, and which categories represent meaningful business exposure?
Seller Performance	Which sellers show combinations of customer and operational underperformance?

Rather than looking for isolated low scores, I focused on patterns that were **consistent, measurable and meaningful at business scale**.

This distinction became important throughout the chapter. A poor metric does not automatically represent a business priority. The number of customers or orders behind that metric also matters.

4.2 Delivery Performance and Customer Satisfaction

Delivery was the first operational factor investigated because the orders dataset contains both the estimated delivery date promised to the customer and the actual date the order was delivered.

Instead of reducing delivery performance to a simple late/not-late flag, delivered orders were grouped according to the extent to which actual delivery differed from the estimated date.

This produced five delivery-performance groups.

Delivery Performance	Orders	Avg. Review Score	% of Orders
Arrived 3+ days early	83,922	4.30	87.10%
Arrived on time / early	6,022	4.11	6.25%
1 to 3 days late	1,856	3.29	1.93%
4 to 7 days late	1,756	2.10	1.82%
7+ days late	2,797	1.70	2.90%

The pattern was clear.

Average customer satisfaction declined at every stage as delivery performance worsened:

4.30 → 4.11 → 3.29 → 2.10 → 1.70

Orders delivered three or more days early recorded an average review score of **4.30**. Orders arriving between one and three days late fell to **3.29**, while orders delayed by seven days or more averaged only **1.70**.

This was not a small difference between two groups. It was a progressive decline across increasingly severe delivery delays.

What the Result Means

The analysis identifies **delivery reliability as the strongest operational satisfaction signal found in this investigation.**

Customers receiving orders within or ahead of the expected delivery window generally reported much stronger satisfaction than customers experiencing delays, particularly longer delays.

There is also an important distinction between **speed and reliability**.

The business lesson is not necessarily that every order needs to arrive as quickly as possible. What appears important from this analysis is whether the actual customer experience meets the delivery expectation established when the order was placed.

That distinction later influenced the recommendation made in Chapter 5.

What the Result Does Not Mean

The analysis is observational.

It demonstrates a strong association between delivery performance and review score, but it does not prove that a delivery delay caused an individual customer to leave a particular rating.

Other aspects of the transaction may also influence the final review.

This distinction is important because the purpose of the analysis is to identify where Olist should investigate and monitor performance, not to claim causal relationships that the dataset cannot establish.

Why This Matters to Olist

Delivery performance provides something particularly valuable from a management perspective: **an operational measure that can be monitored before it becomes a marketplace-wide customer-satisfaction problem.**

Rather than waiting for overall review scores to decline, Olist can monitor delivery reliability at order and seller level and investigate areas where late deliveries become persistent.

This makes delivery performance more than an analytical finding. It becomes a potential **early-warning indicator for customer-experience risk.**

Key Insight 1: Customer satisfaction deteriorates substantially as delivery delays increase. Delivery reliability should therefore be treated as a core customer-experience indicator.

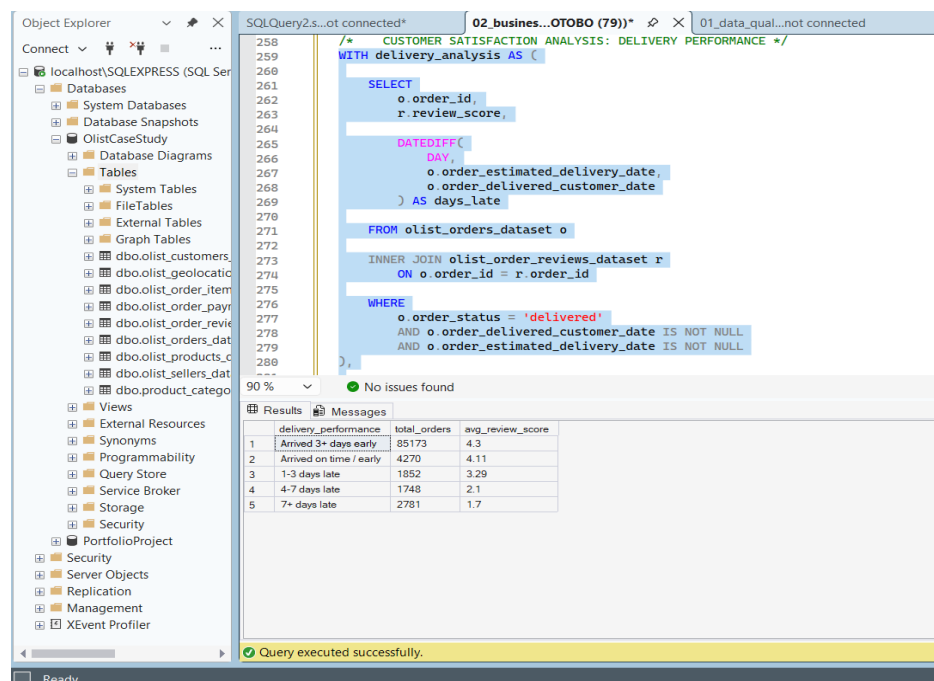


Figure 4. Average customer review score declines progressively as delivery delays increase.

4.3 Product Categories: Separating Severity From Scale

The second part of the analysis investigated whether customer dissatisfaction was distributed evenly across the product catalogue.

It was not.

Some categories recorded noticeably lower average review scores and higher negative-review rates than others.

Examples from the analysis included:

Product Category	Orders	Avg. Review Score	Negative Reviews	Negative Review Rate
moveis_escritorio	1,244	3.52	423	25.42%
telefonica_fixa	209	3.76	58	22.92%
fashion_roupa_masculina	105	3.76	31	25.00%
audio	345	3.83	78	21.73%
casa_conforto	390	3.85	85	19.77%
cama_mesa_banho	9,177	3.92	2,011	18.31%
moveis_decoracao	6,260	3.95	1,484	18.19%
informatica_acessorios	6,499	3.98	1,312	17.10%

At first glance, it would be tempting simply to rank categories by negative-review percentage and identify the highest percentage as the biggest problem. That would be misleading.

4.4 Why Volume Changes the Interpretation

Consider two categories from the results.

fashion_roupa_masculina recorded a **25.00% negative-review rate**, which appears concerning. However, the result is based on only **105 orders**, with 31 negative reviews.

cama_mesa_banho, by comparison, recorded a lower negative-review rate of **18.31%**, but represented **9,177 orders and 2,011 negative reviews**.

The second category therefore represents a much larger number of poor customer experiences despite having a lower negative-review percentage.

This introduced one of the most important concepts in the case study:

Business risk should be evaluated using both severity and scale.

A percentage describes the concentration of a problem.

Volume describes how widely that problem affects the business.

Neither measure is sufficient by itself.

From Ranking to Risk Assessment

This changed how the category analysis was presented.

Rather than producing a simple bar chart ranking categories from worst to best, the Tableau analysis combined:

- **average review score;**
- **negative-review rate;** and
- **order volume.**

Categories with fewer than **500 orders** were filtered from the executive risk visual. This did not mean those categories were considered unimportant or removed from the underlying analysis. The threshold was used to keep the executive view focused on categories where dissatisfaction occurred alongside meaningful marketplace activity.

The full SQL results remained available for deeper investigation.

Why This Matters to Olist

If Olist prioritised categories based only on the highest negative-review percentage, resources could be directed towards relatively small areas of the marketplace while larger sources of dissatisfaction receive less attention.

A better prioritisation method asks:

How poor is the customer experience, and how many customer experiences are affected?

That provides a stronger basis for deciding which categories deserve deeper investigation.

Key Insight 2: Product-category risk is not defined by the worst percentage alone. High-volume categories with elevated dissatisfaction can represent greater customer-experience exposure than smaller categories with more extreme rates.

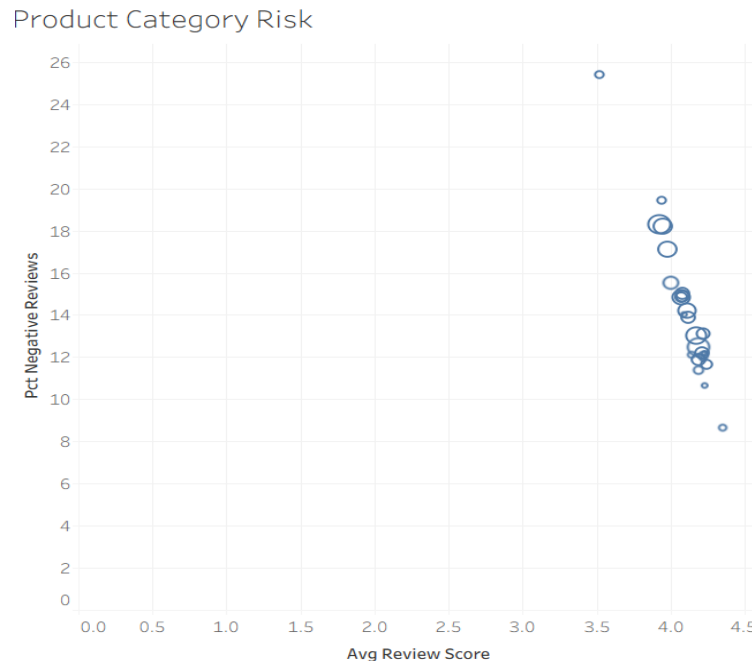


Figure 5. Product-category risk viewed through average review score, negative-review rate and order volume.

4.5 What Category Performance Does Not Tell Us

It would be easy to interpret a poorly performing category as evidence that the products themselves are responsible for customer dissatisfaction.

The analysis does not establish that.

A category-level review pattern may reflect several aspects of the customer experience, including seller performance, fulfilment reliability, delivery performance or characteristics of the products themselves.

The correct business response is therefore **investigation, not immediate attribution**.

A high-risk category should become a starting point for deeper analysis rather than a conclusion about why customers are dissatisfied.

This is another reason the final solution uses a **Category Risk Watchlist** rather than labelling individual categories as inherently problematic.

4.6 Seller Performance: Moving From Marketplace Average to Individual Risk

The final area of investigation examined seller-level performance.

This was particularly important because marketplace averages can hide variation between individual sellers.

A strong overall review score does not mean every seller is providing the same customer experience.

However, seller analysis introduces another problem: **small samples**.

A seller with only a handful of orders could receive one or two poor reviews and appear dramatically worse than a seller handling hundreds of transactions.

To reduce this risk, the seller analysis included only sellers with at least **30 delivered orders**.

This minimum-volume threshold was used to improve the reliability of the comparison, not to suggest that smaller sellers are irrelevant.

4.7 Building a Multi-Metric Seller View

Seller performance was not evaluated using review score alone.

The SQL analysis brought together:

Measure	Purpose
Total Orders	Establish the scale of the seller's marketplace activity
Average Review Score	Measure overall customer satisfaction
Negative Reviews	Identify the volume of particularly poor experiences
Late Delivery Rate	Add operational delivery context
Revenue	Provide commercial context
Seller Location	Support identification and deeper investigation

This produced a more balanced view of seller performance.

The results showed that sellers with meaningful order volumes could still combine relatively low customer ratings with elevated negative-review counts and delivery problems.

For example, one of the lowest-rated sellers in the analysed population had **107 orders**, an average review score of **2.27**, **80 negative reviews** and a **22.43% late-delivery rate**.

Another seller had **184 orders**, an average review score of **2.81**, **93 negative reviews** and a **14.67% late-delivery rate**.

These examples demonstrate why marketplace-wide averages alone are insufficient for seller management.

4.8 Identifying Seller Risk Without Oversimplifying It

As with product categories, the purpose was not to create a list of "bad sellers".

Seller performance is multidimensional.

A seller could have a relatively high late-delivery rate but low order volume. Another could have a moderate dissatisfaction rate but affect substantially more customers. Commercial importance may also differ considerably between sellers.

The Tableau view therefore used:

- **average review score** on one axis;
- **late-delivery rate** on the other;
- **order volume** to determine mark size; and
- additional seller information in the tooltip.

This allows decision-makers to identify sellers where several risk signals occur together rather than relying on a single ranking.

Why This Matters to Olist

The marketplace model means Olist needs visibility not only into overall platform performance but also into the partners contributing to the customer experience.

A seller-level monitoring framework could help distinguish between normal variation and sellers showing persistent weakness across multiple indicators.

The appropriate response would not automatically be seller removal.

Depending on the underlying problem, intervention could involve operational review, seller support, fulfilment improvements, more realistic delivery expectations or continued monitoring.

Key Insight 3: Seller-level analysis reveals operational variation that marketplace averages can conceal. Sellers should be evaluated using multiple performance indicators and sufficient transaction volume before intervention is considered.

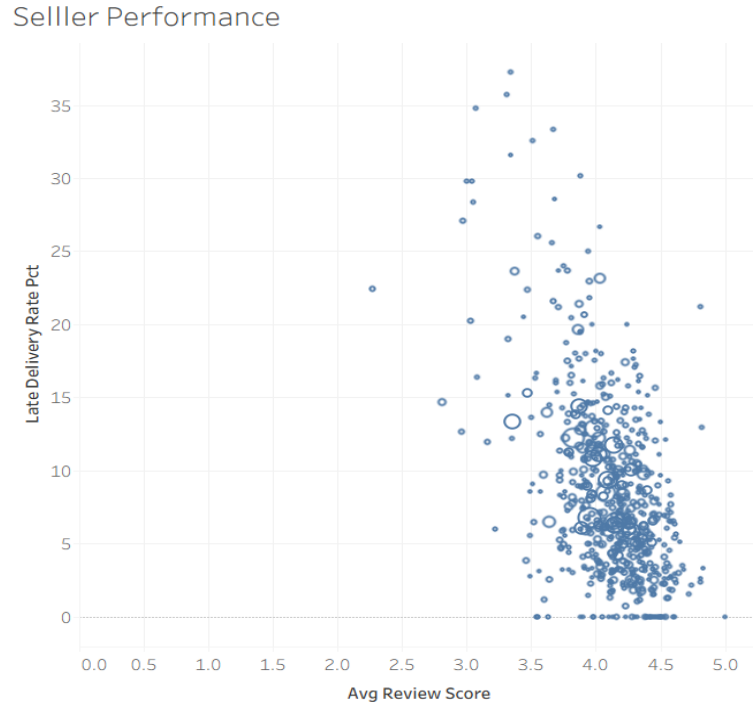


Figure 6. Seller performance evaluated using customer satisfaction, late-delivery rate and order volume.

4.9 Bringing the Three Findings Together

The three areas of analysis are related rather than independent.

The investigation started with customer satisfaction as the outcome and progressively moved deeper into the operational structure of the marketplace.

Finding	What It Revealed	Business Use
Delivery Performance	Satisfaction declines sharply as delivery delays increase	Monitor delivery reliability as a customer-experience signal
Product Category Risk	Dissatisfaction varies across categories and must be interpreted alongside volume	Prioritise categories using severity and scale
Seller Performance	Marketplace averages conceal meaningful seller-level variation	Identify sellers requiring monitoring or deeper operational review

Together, the findings created a clearer view of customer-experience risk.

The analysis did not identify one universal explanation for every negative review. Instead, it identified **where the business has useful signals that can support prioritisation and further investigation.**

4.10 From Insight to a Decision Framework

At this stage, the original business problem could be answered more clearly.

Customer dissatisfaction in the Olist dataset is not distributed randomly across every operational dimension examined.

The strongest relationship identified was between **delivery performance and customer satisfaction**.

Category analysis then showed that dissatisfaction needs to be interpreted in the context of business scale.

Seller analysis showed that marketplace-level performance can conceal individual partners with combinations of weaker customer and operational outcomes.

This led to a simple decision framework:

Signal	Management Question
Delivery reliability deteriorates	Where are orders failing to meet the promised delivery expectation?
Category dissatisfaction increases at meaningful volume	Which categories require deeper operational investigation?
Seller metrics deteriorate across several measures	Which sellers should move from routine monitoring to review?

This is the point where the project moved beyond exploratory analysis.

The findings were now strong enough to determine what the final dashboard needed to communicate and what business actions could reasonably be recommended.

Chapter 4 Outcome

Chapter 4 transformed the validated data into three evidence-based findings.

Delivery reliability emerged as the strongest operational satisfaction signal, with average review scores declining from **4.30 for orders arriving at least three days early to 1.70 for orders arriving seven or more days late**.

Product-category analysis demonstrated that prioritisation requires both **severity and scale**, because the highest dissatisfaction percentage does not necessarily represent the greatest number of affected customer experiences.

Seller-level analysis showed that platform averages can conceal meaningful operational variation and that seller performance is better evaluated through a combination of customer satisfaction, delivery reliability and transaction volume.

These findings answered the analytical questions established in Chapter 1, but analysis alone does not improve a business.

The final step was to decide **what Olist should do with the evidence**.

That became the focus of **Chapter 5: From Insight to Business Action**.

Chapter 5: From Insight to Business Action

5.1 Turning Analysis Into Decisions

The first four chapters moved progressively from defining the customer satisfaction problem to understanding the data, validating its reliability and identifying the strongest patterns in the evidence.

Chapter 5 addresses the final question:

What should Olist do with what the analysis has revealed?

The purpose of this stage is not to repeat the findings from Chapter 4. It is to translate them into priorities that a business team could monitor, investigate and act upon.

Three findings shape the recommendations:

Evidence	Business Implication
Review scores fall substantially as delivery delays increase	Delivery reliability is closely associated with customer experience
Dissatisfaction varies by category, but volume changes its business significance	Category risk requires both severity and scale
Seller performance varies across multiple operational and satisfaction measures	Platform averages can conceal seller-level risk

These findings do not suggest that every negative review has the same cause. Instead, they provide Olist with a more focused framework for deciding **where investigation and operational attention should begin**.

5.2 Executive View of Marketplace Performance

Before moving into specific recommendations, the final dashboard provides context for the scale and overall performance of the marketplace population included in the executive KPI analysis.

Executive KPI	Result
Delivered Orders	95,824
Overall Average Review Score	4.08 / 5
Gross Merchandise Value	R\$15.36M
Platform Late-Delivery Rate	8.93%
Negative-Review Rate	14.79%
Average Freight Cost	R\$19.93

At an aggregate level, an average review score of 4.08 suggests generally positive customer sentiment. However, Chapter 4 demonstrated why the average cannot be interpreted in isolation.

A marketplace can perform well overall while still containing specific delivery experiences, categories and sellers where customer outcomes are considerably weaker.

The executive KPIs therefore provide **context rather than diagnosis**. The analytical views beneath them explain where management attention may be required.

A Note on KPI Populations

Some intermediate calculations produced earlier in the investigation used different filters and analytical populations. For that reason, the executive KPI values above should not be directly mixed with metrics produced from differently filtered queries.

The final executive dashboard uses the dedicated KPI query as its consistent summary population.

This distinction is important because apparently similar metrics can produce different values when calculated at different grains or using different eligibility criteria.

5.3 The Executive Dashboard

The final Tableau dashboard was designed to answer the business problem in a single decision-support view.

Rather than displaying every analysis completed during the project, it contains the measures and visuals most relevant to management action:

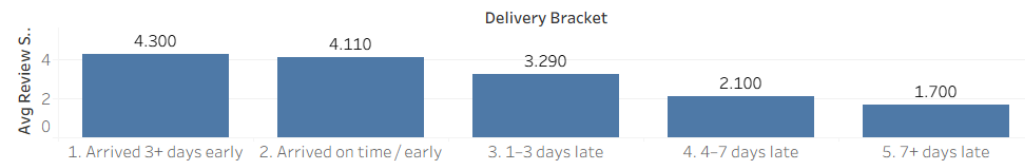
1. **Executive Summary**, providing overall marketplace context.
2. **Delivery Performance**, showing how satisfaction changes with delivery reliability.

3. **Product Category Risk**, highlighting the intersection between dissatisfaction and transaction volume.
4. **Seller Performance**, identifying marketplace partners where multiple risk indicators can be investigated together.

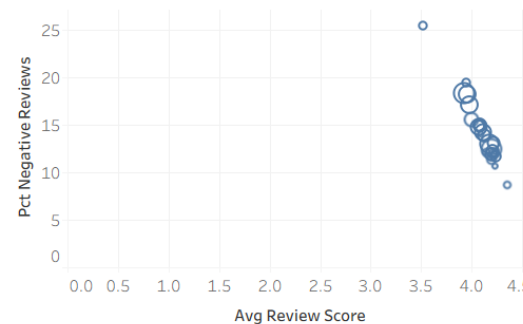
KPI - Executive Summary

Delivered Orders	95,824
Average Review Score	4.080
GMV	R\$15.36M
Late Delivery Rate	8.93%
Negative Review Rate	14.79%
Avg Freight Cost	19.93

delivery performance



Product Category Risk



Seller Performance

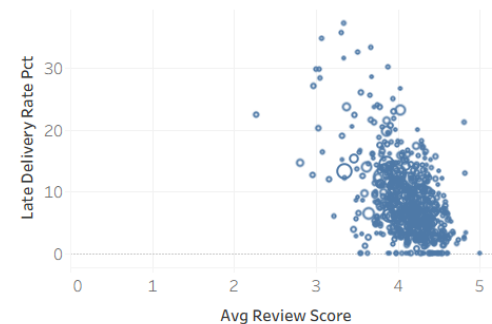


Figure 7. Olist Customer Satisfaction Executive Dashboard, combining marketplace KPIs with delivery, category and seller risk analysis.

Why the Dashboard Was Designed This Way

The dashboard follows the same logic as the investigation.

A stakeholder first sees the overall state of the marketplace, then moves into the strongest operational relationship, followed by two areas where the business can identify more specific sources of risk.

This keeps the dashboard focused on decision support rather than turning it into a collection of every chart produced during the analysis.

5.4 Priority 1: Protect the Delivery Promise

Evidence

Delivery performance produced the strongest satisfaction pattern identified in the case study.

Average review scores declined from **4.30 for orders delivered at least three days early to 1.70 for orders delivered seven or more days late.**

The decline was progressive rather than isolated:

Delivery Experience	Avg. Review Score
3+ days early	4.30
On time / early	4.11
1 to 3 days late	3.29
4 to 7 days late	2.10
7+ days late	1.70

This makes delivery reliability the clearest operational priority emerging from the investigation.

Recommendation

Olist should treat delivery reliability against the promised delivery date as a core customer-experience KPI and establish an exception-monitoring process for persistent delivery underperformance.

The emphasis should be on **reliability**, not simply faster shipping.

If an estimated delivery date is communicated to the customer, consistently meeting that expectation may be more useful operationally than pursuing speed without predictability.

What Olist Should Do

Delivery monitoring should move from a marketplace-wide percentage towards a more targeted exception process.

Management should monitor late-delivery performance by seller and other relevant operational segments, then identify areas where delays persist rather than reacting to isolated late orders.

Orders approaching or exceeding their estimated delivery dates can provide an operational warning signal, while sellers with sustained late-delivery rates can be moved into deeper review.

The purpose is to move from:

"How many orders were late?"

to:

"Where is late delivery becoming a recurring customer-experience risk?"

How Success Should Be Measured

If this recommendation were implemented, management should track whether:

- the platform late-delivery rate declines;
- the share of orders experiencing severe delays declines;
- review scores among delayed orders improve;
- the number of sellers showing persistent delivery underperformance decreases; and
- the gap in satisfaction between on-time and delayed orders narrows.

The historical dataset cannot determine what improvement target Olist should adopt today. Targets should be established using current operational baselines.

5.5 Priority 2: Prioritise Category Risk by Impact, Not Percentage Alone

Evidence

The product-category analysis demonstrated that the category with the most extreme negative-review percentage is not automatically the category representing the greatest customer impact.

For example, `fashion_roupa_masculina` recorded a **25.00% negative-review rate**, but the analysis contained only **105 orders** for that category.

By comparison, `cama_mesa_banho` recorded:

- **9,177 orders**
- **3.92 average review score**
- **2,011 negative reviews**
- **18.31% negative-review rate**

Its negative-review percentage was lower, but considerably more customer experiences were affected.

Recommendation

Olist should introduce a volume-adjusted Category Risk Watchlist that evaluates dissatisfaction using both severity and scale.

Category prioritisation should consider a combination of:

negative-review rate + negative-review volume + average review score + order volume

rather than ranking categories by a single metric.

What Olist Should Do

Categories with meaningful transaction volume and elevated dissatisfaction should be flagged for deeper investigation.

The next step should not be to assume that the product category itself caused the poor experience. Instead, the flagged category should be broken down further to investigate possible contributing factors such as:

- seller mix;
- delivery reliability;
- fulfilment performance; and
- other operational characteristics available to the business.

This creates a two-stage process:

Identify category risk → Diagnose what is driving it

That is more defensible than treating a category-level correlation as a root-cause finding.

How Success Should Be Measured

Management could monitor:

- negative-review rate within watchlist categories;
- total negative reviews generated by high-volume categories;
- average review score within priority categories; and
- movement of categories into or out of the risk watchlist over time.

A successful intervention should improve customer outcomes at meaningful transaction volume, not merely improve an isolated percentage.

5.6 Priority 3: Build a Multi-Metric Seller Monitoring Framework

Evidence

The seller analysis demonstrated that marketplace-level averages can conceal meaningful variation between individual sellers.

To avoid drawing conclusions from very small transaction histories, the analytical comparison was restricted to sellers with at least **30 delivered orders**.

Seller performance was then evaluated using several measures rather than review score alone:

Indicator	What It Contributes
Average review score	Overall customer satisfaction
Negative reviews	Volume of poor customer experiences
Late-delivery rate	Operational reliability
Order volume	Scale and reliability context
Revenue	Commercial context

This allowed sellers showing several risk signals simultaneously to be distinguished from sellers experiencing an isolated poor metric.

Recommendation

Olist should implement a tiered seller-monitoring framework that combines customer satisfaction, delivery reliability, transaction volume and commercial context. Seller management should focus on persistent multi-metric underperformance rather than a single poor review or temporary fluctuation.

Proposed Intervention Framework

A simple operational model could use three stages:

Status	Meaning	Suggested Response
Monitor	Early warning or isolated weakness	Continue tracking performance
Review	Persistent weakness across meaningful volume or multiple indicators	Investigate operational causes and engage seller
Intervention	Sustained multi-metric underperformance	Agree corrective actions and monitor improvement

Intervention should not automatically mean removing a seller from the marketplace.

Depending on the problem identified, action could involve fulfilment support, improved delivery estimates, operational guidance or another corrective plan appropriate to the underlying cause.

How Success Should Be Measured

Management could track:

- the number of sellers moving between monitoring tiers;
- seller late-delivery rates;
- seller average review scores;

- negative-review volumes;
- performance following intervention; and
- the proportion of high-volume sellers remaining in persistent risk status.

This turns seller analysis from a static ranking into an ongoing performance-management process.

5.7 A Combined Customer Experience Monitoring Framework

The three recommendations are strongest when treated as parts of the same system rather than independent initiatives.

A practical management framework could work as follows:

Signal	Trigger	Business Response
Delivery Risk	Persistent or severe lateness	Investigate fulfilment and delivery performance
Category Risk	Elevated dissatisfaction at meaningful volume	Diagnose category-level drivers
Seller Risk	Persistent weakness across multiple indicators	Move seller through Monitor, Review and Intervention process

The dashboard can act as the executive entry point into this framework.

It tells management **where attention is required**. More detailed operational analysis can then determine **why the problem is occurring and what intervention is appropriate**.

This distinction is important.

A dashboard should support prioritisation. It should not pretend to replace operational diagnosis.

5.8 Recommended Order of Action

Not every finding should receive equal priority.

Based on the strength of the evidence in this case study, I would recommend the following sequence:

Priority 1: Delivery Reliability

Begin here because delivery performance showed the clearest and most consistent relationship with customer satisfaction.

Priority 2: Seller Monitoring

Use seller-level performance to identify where delivery and customer-experience problems may be concentrated among marketplace partners.

Priority 3: Category Investigation

Use the volume-adjusted category framework to identify commercially meaningful areas of dissatisfaction and investigate the operational factors behind them.

This prioritisation reflects the strength and actionability of the evidence rather than treating all three analytical findings as equally important.

5.9 What I Would Analyse Next

The current project identifies where important customer-experience patterns exist. A subsequent phase should move closer to root-cause analysis.

The most useful next investigations would include:

Seller and delivery interaction: Determine how much of seller-level dissatisfaction overlaps with delivery underperformance.

Category and delivery interaction: Test whether high-risk categories also experience unusually poor delivery performance.

Geographic patterns: Investigate whether delivery risk is concentrated across particular customer-seller routes or regions.

Time trends: Determine whether satisfaction and delivery performance improved or deteriorated across the historical period.

Review text analysis: Where review comments are sufficiently available and suitable for analysis, text analysis could help identify recurring themes behind low scores.

These are deliberately presented as **future analyses**, not conclusions from the current project.

That distinction protects the integrity of the case study.

5.10 Limitations and Responsible Interpretation

The analysis provides useful evidence, but its findings should be interpreted within the boundaries of the dataset and methodology.

Limitation	Impact on Interpretation
Historical dataset	The data covers 2016 to 2018 and should not be presented as Olist's current performance.
Observational analysis	Associations should not automatically be interpreted as causal relationships.
Missing reviews	646 delivered orders did not contain a review and therefore could not contribute to review-based satisfaction analysis.
Missing categories	610 products, approximately 1.85%, lacked product-category information.
Seller threshold	Sellers with fewer than 30 delivered orders were intentionally excluded from the main seller comparison.
Review score as satisfaction measure	A review may reflect several parts of the customer experience and cannot isolate one cause by itself.
Category interpretation	Poor category performance does not prove that the products themselves caused dissatisfaction.

These limitations do not invalidate the findings. They define **how far the findings can reasonably be taken**.

That distinction is important in business analytics. A strong recommendation should be confident where the evidence is strong and cautious where additional investigation is required.

5.11 Final Conclusion: From Business Problem to Decision Support

This case study began with a business problem rather than a dataset.

Olist needed a clearer way to understand what sits behind customer satisfaction, where dissatisfaction is concentrated and which areas should receive operational attention.

Chapter 1: Defining the Problem

The investigation first established that an overall review score was not enough to support those decisions. Customer experience needed to be examined in the context of the wider transaction, particularly delivery performance, product categories and individual sellers.

The business requirement was therefore translated into focused analytical questions before any conclusions were attempted.

Chapter 2: Understanding the Data

The next stage established how Olist's relational datasets represented that customer journey.

Orders could be connected to reviews, delivery timestamps, products and sellers. Understanding the grain and relationships between those tables provided the analytical structure required to investigate the business questions without distorting the underlying metrics.

Chapter 3: Establishing Trust

Before using those relationships to produce insights, the evidence was validated.

The project investigated table completeness, order statuses, historical coverage, missing delivery dates, review coverage and product-category completeness. Import and datatype issues were corrected rather than ignored.

This established that the dataset was suitable for analysis while preserving clear limitations around how the results should be interpreted.

Chapter 4: Finding the Patterns

The analysis then identified three important patterns.

Delivery reliability produced the strongest relationship with customer satisfaction. Average review scores declined progressively from **4.30 for orders delivered at least three days early to 1.70 for orders arriving seven or more days late**.

Product-category analysis showed that dissatisfaction cannot be prioritised using percentage alone. Business impact depends on both **severity and scale**.

Seller analysis demonstrated that marketplace averages can hide individual operational risk and that seller performance is better understood through several indicators rather than one metric.

Chapter 5: Turning Evidence Into Action

Those findings were translated into three management priorities:

Protect the delivery promise.

Prioritise category risk using severity and scale.

Monitor sellers using multiple operational and customer-experience indicators.

The final Tableau dashboard brings those priorities together in a single decision-support view, while the recommended monitoring framework provides a practical route from identification to investigation and intervention.

Final Takeaway

The strongest conclusion from this case study is not simply that late orders receive lower ratings or that some sellers and categories perform worse than others.

The more useful conclusion is that **customer dissatisfaction leaves operational signals that can be identified before they disappear inside marketplace averages.**

Delivery reliability provides the strongest signal identified in this analysis. Category volume provides the context needed to prioritise dissatisfaction intelligently. Seller-level monitoring provides a way to locate operational risk more precisely.

Together, these findings provide Olist with a framework for moving from:

measuring customer satisfaction after the fact

to:

using customer and operational data to identify where improvement should begin.

The completed project therefore follows the full analytical journey established at the outset:

Business Problem → Data Understanding → Data Validation → Evidence → Insight → Decision Support

The value of the analysis is not the dashboard alone. It is the chain of evidence connecting a business question to a defensible recommendation.